

IAL Biology Unit 5 (WBI15/01) — Complete A* Study Note

Respiration, Internal Environment, Coordination and Gene Technology

Built entirely from analysis of 20 past papers and their mark schemes, June 2019 → June 2026.

Everything below is traceable to what examiners actually wrote. Where a mark scheme states an **ACCEPT**, **ALLOW**, **IGNORE** or **DO NOT ACCEPT** ruling, it is quoted verbatim — those rulings are where the marks are decided. This note is designed to be the only thing you need.

0. THE PAPER — AND THE 20-MARK QUESTION THAT DECIDES IT

Code	WBI15/01
Time	1 hour 45 minutes
Marks	90
Structure	Questions 1–7 on specification content (~70 marks), then Question 8 on the pre-released scientific article (20 marks)
Asterisked (*) questions	1–3 per paper, levels-based marking ; older papers also carry QWC — <i>“emphasis is clarity of expression”</i>
Maths	Minimum 9 marks at Level 2 or above
Synoptic	May draw on Units 1 and 2
Equipment	Scientific calculator, ruler, HB pencil

Pace: 90 marks in 105 minutes $\approx$ **1.17 min/mark**. Work backwards from the clock: **Question 8 needs 22–25 minutes**, so Questions 1–7 must be done in about 80.

⚠ Question 8 is a fifth of the paper, and it is the same every session

Every WBI15 paper ends with a **20-mark question on a scientific article released months in advance**. You are given a **clean, unannotated copy** in the exam, so all preparation must already be in your head.

Articles used so far: *Venom: the secrets of nature's killers* (Jun 2019) · *First in Fly: Drosophila* (Oct 2019) · *National Geographic* (Oct 2020) · *Education in Chemistry* (Mar 2021) · *Journal of...* (Apr 2021) · *Rethinking Alzheimer's* (Oct 2021) · *Science Direct* (Jan 2022) · *StatPearls* (Jun 2022) · *How we perceive our environment* — the 2021 Nobel Prize on **TRPV1, TRPM8 and Piezo channels** (Oct 2022) · *New Scientist* (Oct 2023, Jun 2024) · *Clinical Nutrition* (Jan 2024) · *Alzheimer's* (Oct 2024) · *New Scientist* on hydration and drinks (Jan 2025) · *Risk of heart disease* (Jun 2025) · *Popular Science* (Oct 2025) · *Nature* (Jan 2026) · *New Scientist* (Jun 2026).

The critical insight from the mark schemes: the article questions are marked almost entirely against **ordinary specification biology**. Look at what the June 2019 *Venom* mark scheme actually credits (Q7g–j):

*"calcium ions (entering presynaptic neurone) **bind to vesicles containing neurotransmitter**" · "causing vesicles to **move to / fuse with** presynaptic membrane" · "releasing contents into synapse" — "ALLOW release contents by **exocytosis**" · "serotonin is a neurotransmitter" · "binds to **pain receptors**" — "ALLOW binds to **post-synaptic receptors/membrane**" · "on **sensory neurones**" · "impulses pass to the **brain/CNS**" · "using **fMRI**" · "**compare images from control and stung individuals**"*

That is Topic 8 synapse biology, straight out of §22 of this note, wearing a snake-venom costume. **Roughly two-thirds of the 20 marks are specification content applied to the article's context**. §40 is the full preparation method.

Topic weighting across the 20 papers

Topic	Marks/paper	Appears in
Pre-released article	20	Every paper
Nervous system: neurones, action potentials, synapses	12–20	Every paper
Respiration (four stages, RQ, mitochondria)	10–18	Every paper
Kidney: ultrafiltration, reabsorption, ADH	8–15	Every paper
Muscles, joints and movement	6–14	~90%
Heart, exercise, cardiac output, ventilation	6–14	~85%

Brain, imaging, Parkinson's, drugs	5–14	~85%
Plant responses: phytochrome, IAA, gibberellin	4–10	~80%
Eye, rods and cones	4–10	~75%
Homeostasis, feedback, thermoregulation	3–10	~70%
Gene technology: recombinant DNA, microarrays	4–12	~55%

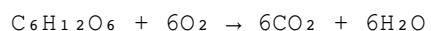
Nervous system + respiration + kidney = 30–50 of the 70 non-article marks. Those three are the spine.

The four things the mark schemes reward that students miss

1. **The precise word, not the near-synonym.** *"IGNORE nerve impulses"* for the SAN. *"IGNORE hydrogen"* where **H⁺/proton** is needed. *"ALLOW arteriole NOT artery."* *"IGNORE phosphorylation makes ATP."*
2. **The mechanism, not the outcome.** *"IGNORE no depolarisation"* — you must say **reducing depolarisation below the threshold**.
3. **Data quoted with figures.** "Describe the data" marks require numbers from the graph or table.
4. **Linked chains.** Levels-based questions cap unlinked lists at Level 2.

PART A — TOPIC 7: RESPIRATION, MUSCLES AND THE INTERNAL ENVIRONMENT

1. ATP and the overall reaction



The specification's framing, which mark schemes echo: respiration involves **splitting the respiratory substrate to release carbon dioxide**, and **reuniting the hydrogen with atmospheric oxygen**, releasing a large amount of energy.

Why respiration proceeds in many small steps: it is a **stepped process**, each step

catalysed by a specific intracellular enzyme; releasing energy in **small amounts** means it can be **captured as ATP** rather than **lost as heat**, and each stage can be **controlled**.

Why ATP rather than direct use of energy: ATP is **small and soluble**, so it can be **moved to where it is needed**; it releases energy in **small usable amounts**; and it is the **immediate energy source** for biological processes.

2. Glycolysis and anaerobic respiration

Location: the cytoplasm. Occurs in both aerobic and anaerobic respiration. No oxygen required.

1. PHOSPHORYLATION Glucose (6C) + 2 ATP → hexose biphosphate (6C)
2. LYSIS hexose biphosphate → 2 × triose phosphate (3C)
3. OXIDATION 2 triose phosphate → 2 PYRUVATE (3C)
 + 2 REDUCED NAD + 4 ATP (substrate-level phosphorylation)

NET per glucose: 2 ATP · 2 reduced NAD · 2 pyruvate

Why glucose is phosphorylated first: it makes the molecule **more reactive / less stable**, so it can be split, and it **traps the glucose in the cell**.

🔗 **Substrate-level phosphorylation:** ATP formed by the **direct transfer of a phosphate group from a substrate to ADP**, not by chemiosmosis. The June 2019 Q5b(iii) mark scheme credits *"(some ATP is produced) by **substrate level phosphorylation**" — "ALLOW some ATP is also produced directly."*

2.1 The NAD trap

June 2019 Q5a(ii) — the mark scheme's answer is **"NAD accepts electrons in glycolysis and in the Krebs cycle"**, and it rejects the distractors with two separate points:

*"it is **NAD not reduced NAD** that accepts electrons" "NAD is an electron acceptor in **both glycolysis and Krebs cycle**"*

⚠ **Reduced NAD is the product, not the acceptor.** And NAD is reduced in glycolysis, the link reaction *and* the Krebs cycle.

2.2 Anaerobic respiration

ANIMALS: Pyruvate + reduced NAD → LACTATE + NAD (lactate dehydrogenase)

YEAST/PLANTS: Pyruvate → ethanal + CO₂ ; ethanal + reduced NAD → ETHANOL + NAD

The purpose, and the mark scheme's own phrasing (June 2019 Q5b(ii)): the reaction **regenerates NAD** so that **glycolysis can continue** and ATP can still be made by **substrate-level phosphorylation** — "*oxidative phosphorylation to regenerate NAD*" is the aerobic counterpart it contrasts with.

Yield: only **2 ATP** anaerobically versus ~32 aerobically, because the substrate is **not fully oxidised** — lactate and ethanol still contain much chemical energy.

2.3 The lactate/glycogen * question — a model of how these are marked

June 2019 Q5b(i), a 6-mark QWC question — "*emphasis is clarity of expression.*" The mark scheme's eight points, paired as description and explanation:

1. **rate of lactate production decreased** with duration of activity
2. **rate of ATP production decreased** with duration of activity
3. **glycogen used increases** with duration of activity
4. as **glycogen is converted to glucose**
5. **glucose is converted to lactate**
6. after **120 seconds glycogen used levels off**
7. **anaerobic respiration decreases or stops**
8. because **glycogen has been completely used up**

Read the structure: points 1, 2, 3 and 6 **describe the data with figures**; points 4, 5, 7 and 8 **explain them**. That alternating pattern is what reaches the top level.

2.4 The fate of lactate and the oxygen debt

Lactate **lowers the pH**, which can **denature enzymes** and causes **muscle fatigue**. It **diffuses into the blood** and is transported to the **liver**, where — with oxygen available — it is **oxidised back to pyruvate** and either enters the **Krebs cycle** or is **converted back to glucose/glycogen**.

The mark scheme (June 2019 Q5) credits any of these as "another correct requirement for oxygen":

"repay oxygen debt / re-oxygenate myoglobin / aerobic respiration of breathing muscles / increased metabolic rate"

Glycolysis	Cytoplasm
Link reaction	Mitochondrial matrix
Krebs cycle	Mitochondrial matrix
Oxidative phosphorylation	Inner mitochondrial membrane (cristae)

4. Oxidative phosphorylation and chemiosmosis

4.1 The 5-mark chain, exactly as the mark scheme lists it

Oct 2019 Q3b — the six creditable points:

1. *"reference to **chemiosmosis**"*
2. *"**use / release energy from electrons**" — "ACCEPT reference to **transfer of electrons along ETC**"*
3. *"**protons moved through the inner membrane / into the intermembrane space**"*
4. *"reference to **ATP synthase / stalked particles**" — "ACCEPT ATP synthetase / ATP-ase"*
5. *"**formation of phosphate bond between phosphate in ADP and inorganic phosphate**" — "IGNORE phosphorylation makes ATP"; "ACCEPT $ADP + P_i \rightarrow ATP$ "*
6. *"by the **movement of protons from intermembrane space into matrix**"*

⚠ **Point 5 is the mark most often thrown away.** Writing "ATP is made by phosphorylation" is explicitly ignored. You must state that a **bond is formed between ADP and inorganic phosphate**, and you must **name ATP synthase**.

4.2 The electron transport chain diagram question

Oct 2019 Q3a(ii) — completing boxes in a flow diagram. The mark scheme's answers, with one sharp rejection:

- first box **reduced NAD / NADH / NADH₂**; second box **NAD⁺ / NAD** • middle box **H⁺ / hydrogen ion / proton** — *"IGNORE hydrogen / H₂ / H"* • fourth box **oxygen / ½O₂ / O**; fifth box **water / H₂O**

⚠ **"Hydrogen"** alone is ignored. Write **H⁺, hydrogen ion** or **proton**.

Why oxygen is essential: it is the **final electron acceptor**, combining with electrons and

protons to form **water**. Without it, the final carrier **cannot pass on its electrons**, the chain **backs up**, reduced NAD and FAD **cannot be reoxidised**, and the **link reaction and Krebs cycle stop**.

4.3 The ATP yield

Source	ATP
Glycolysis (substrate-level)	2
Krebs (substrate-level)	2
10 reduced NAD × ~2.5	~25
2 reduced FAD × ~1.5	~3
TOTAL	~32 per glucose

Reduced NAD yields more than reduced FAD because it delivers electrons **earlier in the chain**, so **more protons are pumped**.

Why the actual yield is lower: ATP is used to **actively transport pyruvate, ADP and phosphate into the mitochondrion**, and the **proton gradient leaks**.

4.4 Mitochondrial structure

Structure	Function
Inner membrane folded into cristae	Large surface area for ETC carriers and ATP synthase
Intermembrane space	Small volume , so the proton gradient builds rapidly
Matrix	Link reaction and Krebs enzymes; mitochondrial DNA and 70S ribosomes
Many mitochondria	In cells with high energy demand — muscle, PCT epithelium, sperm midpiece

5. Respiratory quotient

$$RQ = \text{volume of CO}_2 \text{ produced} \div \text{volume of O}_2 \text{ consumed}$$

Substrate	RQ
Carbohydrate	1.0
Protein	~0.9
Lipid	~0.7
Anaerobic respiration occurring	> 1.0

Why lipid gives a lower RQ: lipids contain **proportionally less oxygen and more hydrogen**, so **more oxygen is needed** to oxidise them.

The exercise question (Jun 2023 Q5b(ii)) — 2 marks, and note the two things the scheme ignores:

• *“the **respiratory quotient decreased** (during the exercise)”* • *“(switching to) **lipid / fat / triglyceride being used** (rather than carbohydrate)”* — *“accept suitably named compounds”* *“ignore ref to anaerobic respiration occurring”* • *“ignore ref to change in CO₂ production”*

⚠ **Do not explain a falling RQ by anaerobic respiration.** A **rising** RQ above 1 indicates anaerobic respiration; a **falling** RQ means a **switch to lipid**.

Worked from an equation (Jan 2024 Q1): $C_{18}H_{34}O_2 + 25.5 O_2 \rightarrow 18 CO_2 + 17 H_2O$, so

$RQ = 18 \div 25.5 = 0.71$ — **a lipid**.

6. The respiration practicals

6.1 Redox indicator with yeast

Methylene blue and DCPIP are artificial hydrogen acceptors. During respiration, **dehydrogenases remove hydrogen**; the indicator **accepts it and is reduced**, turning **blue** → **colourless**.

rate of respiration $\propto 1 \div$ time to decolourise

Method marks: fixed volume of yeast suspension and indicator; **water bath** at a set temperature; start a clock and time the **colour change**; compare with a **colour standard** so the end point is judged consistently; repeat and take a **mean**. **Control: boiled (killed) yeast** — the indicator should **not** decolourise, showing the change requires **living yeast with**

active enzymes. ⚠ **Do not shake the tube** — dissolved oxygen **re-oxidises the indicator**, turning it blue again. **Limitation:** the end point is **subjective**; use a **colorimeter** to improve it.

6.2 The respirometer

Organisms in a sealed tube with **soda lime or KOH to absorb CO₂**, connected to a **manometer**. Because CO₂ is absorbed, any volume change is due **only to oxygen uptake**, and the fluid moves **towards the organisms**.

$$\text{volume of O}_2 = \text{distance moved} \times \pi r^2 \qquad \text{rate} = \text{volume} \div \text{time (per gram)}$$

For RQ — the two-run method:

RUN 1 WITH soda lime: change = O₂ CONSUMED
 RUN 2 WITHOUT soda lime: change = O₂ consumed - CO₂ produced
 CO₂ produced = O₂ consumed - net change from run 2 ; RQ = CO₂ ÷ O₂

The marks: the **control tube of glass beads** of equivalent volume compensates for **changes in temperature and atmospheric pressure**; a **water bath** keeps temperature constant; **equilibrate** before reading; divide by the **mass of organisms**. **Ethics:** use **invertebrates**, keep them in for the **minimum time**, keep soda lime from touching them, and **return them unharmed**.

7. Muscles, joints and movement

7.1 The tissues

Tissue	Role — and the property that matters
Tendon	Attaches muscle to bone ; inelastic , so all the force is transmitted to the bone
Ligament	Attaches bone to bone ; slightly elastic , so it allows movement but prevents dislocation
Cartilage	Reduces friction and absorbs shock
Synovial fluid	Lubricates the joint

🔑 **Antagonistic pair:** two muscles working in **opposite directions** — the **flexor** (biceps) bends the joint, the **extensor** (triceps) straightens it. Needed because **muscles can only pull, not push**.

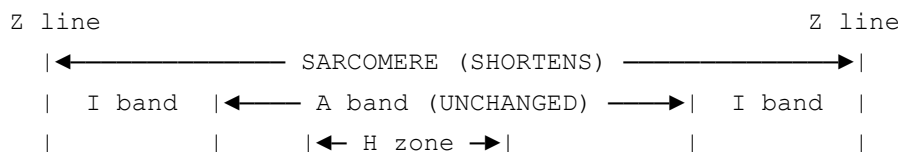
Tendon grafts (Jun 2019 Q2a) — the mark scheme, 3 marks:

• *"a donor tendon may be recognised as being foreign" — "ALLOW will not have the same antigens"* • *"results in an immune response"* • *"leading to rejection of the graft/repair/tendon" — "ALLOW damaging the graft"*

7.2 The sarcomere — the MCQ that catches everyone

Mar 2021 Q4a — the mark scheme's rejections are the teaching:

Structure **X = myosin** — *"A is not correct because the structure labelled X is made from myosin not actin"*, and likewise not tropomyosin, not troponin. The band that **changes in length is Z** — *"W does not change in length during contraction... X does not change... Y does not change."*



Region	During contraction
Sarcomere	SHORTENS
A band	UNCHANGED
I band	SHORTENS
H zone	SHORTENS

⚠ **The A band never changes**, because the **myosin filaments do not shorten** — they **slide past** the actin.

7.3 The sliding filament theory — the eight-step chain

1. An **action potential** travels along the **sarcolemma** and down the **T-tubules**.
2. The **sarcoplasmic reticulum** releases **calcium ions**.
3. **Ca²⁺** binds to **troponin**, which **changes shape**.

4. This moves tropomyosin, exposing the myosin-binding sites on actin.
5. The myosin head binds to actin, forming a cross-bridge.
6. The head changes angle — the power stroke — pulling actin towards the centre; ADP and Pi are released.
7. ATP binds to the myosin head, causing the cross-bridge to detach.
8. ATPase hydrolyses the ATP, and the energy recocks the head, ready to bind again.

Relaxation: Ca^{2+} is actively pumped back into the sarcoplasmic reticulum, so tropomyosin recovers the binding sites.

△ **The two roles of ATP:** detachment of the head (ATP binding) and recocking it (ATP hydrolysis) — plus pumping Ca^{2+} back.

Muscle fatigue (Oct 2024 Q3): ATP and phosphocreatine are depleted; anaerobic respiration produces lactate, which lowers the pH and may inhibit enzymes and interfere with Ca^{2+} binding.

7.4 Slow twitch versus fast twitch

	Slow twitch	Fast twitch
Colour	Red	White
Mitochondria	Many	Few
Capillaries	Many	Few
Myoglobin	High (stores oxygen)	Low
Respiration	Aerobic	Anaerobic
Glycogen / phosphocreatine	Less	More
Contraction	Slow, sustained	Fast, powerful
Fatigue	Resistant	Fatigues quickly
Used for	Endurance, posture	Short bursts

Applied: an endurance athlete has more **slow twitch** fibres — more mitochondria and capillaries mean more **aerobic ATP**, so **less lactate** and **less fatigue**.

8. The heart

8.1 Myogenic — the mark scheme's exact definition

Jun 2019 Q1a(i): the correct answer is *"contraction initiated in a muscle cell"*, and the rejections are instructive:

*"B is not correct because contraction does **not start in a nerve cell**" "C is not correct because **myogenic is the process of contraction and not nerve impulses**" "D is not correct because myogenic is the process of contraction and not nerve impulses **and occurs in muscle cells**"*

8.2 The conduction sequence — with the ruling that costs marks

Jun 2019 Q1c, 3 marks. The mark scheme's five points:

1. *"receives impulses from cardiac control centre" — "ALLOW **cardiovascular centre / medulla (oblongata)**"*
2. *"acts as the **pacemaker**"*
3. *"produces **waves of excitation / impulses / waves of depolarisation**" — "IGNORE **nerve impulses**"*
4. *"(which) **spread across atria / start contraction of atria / triggering atrial systole**"*
5. *"travels to the **AVN**" — "ALLOW **excites the AVN**"*

△ **"Nerve impulses" is explicitly ignored** — the heart is myogenic. Write **waves of excitation** or **waves of depolarisation**.

The full pathway:

SAN (right atrium, the pacemaker) → wave spreads across **both atria** → **atrial systole** → a band of **non-conducting fibrous tissue** prevents direct spread → **AVN**, which **delays** the wave by ~0.1 s **so the atria empty and the ventricles fill** → **bundle of His** through the septum to the **apex** → **Purkyne fibres** up the ventricle walls → ventricles contract **from the apex upwards, pushing blood up and out.**

8.3 Cardiac output — the mark scheme's wording

Jun 2019 Q1a(iii): *"cardiac output is the volume of blood ejected from the left ventricle in each beat multiplied by the heart rate."*

$$\text{cardiac output} = \text{heart rate} \times \text{stroke volume}$$

And Q1a(ii): *"the **left ventricle contracts with the greatest force**"* — because it pumps blood to the **whole body**, so its wall is the thickest.

The ECG calculation (Jun 2019 Q1b(i)) — the mark scheme shows both steps:

$$(11.7 \times 0.2) \div 3 = 0.78 \text{ seconds per beat}$$

$$60 \div 0.78 = 76.9 \text{ beats per minute}$$

$$\text{"ALLOW } (11.8 \times 0.2) \div 3 = 0.787\text{"}$$

$$\text{"ALLOW 75.9 to 77"}$$

And Q1b(ii), when heart rate increases: *"the **distance between peaks would be shorter**"* — *"ALLOW **more frequent peaks**"; "IGNORE **more peaks**."*

8.4 The ECG trace

Feature	Represents
P wave	Atrial depolarisation
QRS complex	Ventricular depolarisation
T wave	Ventricular repolarisation

Abnormalities: **tachycardia** (>100 bpm at rest), **bradycardia** (<60), **ectopic beat**, **fibrillation** (chaotic, no effective pumping), **heart block** (P wave not followed by QRS).

8.5 Control of heart rate and ventilation during exercise

Why both increase: respiring muscles need **more oxygen and glucose** and produce **more CO₂**, which must be removed, so **aerobic respiration can continue** and anaerobic respiration and fatigue are delayed.

Heart rate: the **cardiovascular control centre in the medulla oblongata** receives information from **chemoreceptors** (falling blood pH / rising CO₂), **baroreceptors** (blood pressure) and **stretch receptors in muscles and joints**. It sends impulses via the **sympathetic (accelerator) nerve to increase** rate and the **parasympathetic (vagus) nerve to decrease** it.

Ventilation: the **ventilation centre in the medulla** responds to **chemoreceptors detecting rising CO₂ / falling pH**, sending more frequent impulses to the **diaphragm and intercostal muscles**.

△ It is the rise in CO₂, not the fall in oxygen, that is the main stimulus.

Blood redistribution: **vasodilation** to skeletal muscle, heart and skin; **vasoconstriction** to

gut and kidneys.

$$\text{minute ventilation} = \text{tidal volume} \times \text{breathing rate}$$

Adrenaline (fight or flight): increases **heart rate and stroke volume**; **vasodilation to muscle, vasoconstriction to gut and skin**; increases **ventilation**; causes **glycogenolysis**, raising blood glucose; **dilates the pupils**; raises **metabolic rate**.

8.6 The spirometer

A chamber of oxygen over water, with a **mouthpiece, nose clip** and **soda lime to absorb exhaled CO₂**.

Measurement	From the trace
Tidal volume	height of one normal breath
Breathing rate	peaks per minute
Minute ventilation	tidal volume × breathing rate
Vital capacity	maximum inspiration to maximum expiration
Oxygen consumption	the downward slope of the mid-line , because CO ₂ is absorbed while O ₂ is used

Safety marks: **sterilise or use a disposable mouthpiece**; only **medically fit subjects**; **exclude asthmatics**; ensure the **soda lime is fresh** (exhausted soda lime lets CO₂ accumulate — dangerous and invalidating); check there is **enough oxygen**; **stop if the subject feels dizzy**.

9. Homeostasis, feedback and thermoregulation

🔑 **Homeostasis:** the maintenance of a **constant internal environment** within **narrow limits** around a **set point** — a **dynamic equilibrium**.

🔑 **Negative feedback — the mark-scheme definition (Jun 2019 Q3a):** *“the **sensors detect increases and decreases** in internal body conditions and the **effectors reverse the detected change**.”*

The four-component model:

STIMULUS → RECEPTORS detect the change → COORDINATOR → EFFECTORS REVERSE it
→ conditions return to the set point → the response is switched off

Positive feedback amplifies the change — the **influx of sodium ions during an action potential, oxytocin in labour, blood clotting.**

Thermoregulation — the 4-mark chain, with two rulings

Jun 2019 Q3b:

1. *"thermoreceptors detect an increase in (core) temperature" — "IGNORE location of thermoreceptors"; "ALLOW thermosensors / temperature receptors"*
2. *"send impulses to the **hypothalamus / thermoregulatory centre / heat loss centre**" — "IGNORE medulla (oblongata)"*
3. *"sends impulses to **effector organs**" — "e.g. **sweat glands / arterioles / erector (hair) muscles**"*
4. *"to **increase heat loss**"*
5. *"returns **body temperature to set value**" — "ALLOW temperature returns to normal"*

△ **The medulla oblongata is explicitly ignored here** — thermoregulation is the **hypothalamus**. (The medulla controls heart rate and ventilation.)

The reverse case (Mar 2021 Q3): *"impulses travel to the **thermoregulatory centre in the hypothalamus**" + "description of consequence that would lead to **increased body temperature**" — "e.g. **reduce sweating / increase metabolic rate / increased respiration.**"*

Too hot	Too cold
Vasodilation of skin arterioles → more heat lost	Vasoconstriction → less heat lost
Sweating — evaporation uses latent heat	Shivering — respiration releases heat
Hairs lie flat	Hairs erect , trapping insulating air
Metabolic rate decreases	Metabolic rate increases

Why homeostasis matters: enzymes have an **optimum temperature and pH**; a constant **water potential** stops cells shrinking or bursting; the organism stays **independent of external fluctuations**.

10. The kidney

10.1 Gross and microscopic structure

Gross: renal artery in, renal vein out, ureter to the bladder; **cortex** (outer), **medulla** (inner), **pelvis**, **capsule**.

Nephron part	Location	Function
Bowman's capsule + glomerulus	Cortex	Ultrafiltration
Proximal convoluted tubule	Cortex	Selective reabsorption
Loop of Henle	Medulla	Builds the concentration gradient
Distal convoluted tubule	Cortex	Fine adjustment; ADH-sensitive
Collecting duct	Medulla	Water reabsorption under ADH control

10.2 Ultrafiltration — 2 marks, three rulings

Jan 2021 Q2b, on how urea enters the filtrate:

• *"(urea) forced out by high pressure (of the blood)"* • *"caused by afferent blood vessel greater diameter than efferent blood vessel in the glomerulus"* — **"ALLOW arteriole NOT artery"** • *"through pores in the (basement) membrane"* — **"ALLOW reference to podocytes / glomerular fenestrations"** **"IGNORE any other mechanism other than ultrafiltration"**

⚠ **Three traps in one 2-mark question:** say **arteriole** (not artery); name the **basement membrane pores**; and do not drift into active transport or diffusion.

The three filtration layers: **capillary endothelium** (fenestrations) → **basement membrane** (the actual filter) → **podocytes** (filtration slits).

Filtered: water, **glucose**, **amino acids**, **urea**, ions, vitamins, hormones. **Not filtered:** **blood cells and platelets** (too large) and **plasma proteins** (too large for the basement membrane). Protein or blood in urine indicates **glomerular/basement-membrane damage**.

10.3 Selective reabsorption in the PCT

Jun 2021 Q4d — the mark scheme wants quantitative reference:

• *"proximal tubule has role in selective reabsorption"* • *"named relative*

concentration change by reabsorption" — "e.g. 100 % of glucose" • "comment on protein channels for reabsorption" • "named relative concentration change by secretion" — "e.g. 70× increase for urea"

⚠ **Quote a figure from the data.** A qualitative answer will not reach full marks.

The mechanism:

Sodium ions are actively transported out of the PCT cell into the blood by the **Na⁺/K⁺ pump**, lowering Na⁺ inside the cell. Sodium then **diffuses in from the filtrate through co-transporter proteins, carrying glucose or amino acids with it**. Glucose passes into the blood by **facilitated diffusion**. Removing solutes **lowers the water potential of the blood**, so **water follows by osmosis**.

PCT cell adaptations: **microvilli** (large surface area) • **many mitochondria** (ATP for active transport) • **many carrier and channel proteins** • **tight junctions** • **folded basal membrane**.

Glucose in urine occurs only when the **carrier proteins are saturated** (the renal threshold is exceeded) — hence its use as a sign of **diabetes**.

10.4 The loop of Henle — the 4-mark chain

Mar 2022 Q4b — the mark scheme's four points:

• "flow of filtrate in tubules in opposite direction" (counter-current) • "because sodium ions actively transported out of the ascending limb" • "therefore water out from descending limb (and solutes from ascending limb)" • "due to increase in concentration gradient between tubular fluid and interstitial space"

The full explanation:

1. **Na⁺ and Cl⁻ are actively transported out of the ascending limb** into the medulla; the ascending limb is **impermeable to water**.
2. This makes the **water potential of the medulla very low**, and **progressively lower towards the tip**.
3. The **descending limb is permeable to water but not ions**, so as filtrate descends through the increasingly concentrated medulla, **water leaves by osmosis**.
4. Because the limbs run in **opposite directions**, a gradient is **maintained along the whole length** — this **multiplies it**.
5. The **collecting duct passes back down through this gradient**, so — if ADH has made it permeable — **water leaves by osmosis**, giving **concentrated urine**.

The desert-adaptation question (Jan 2021 Q2c) — the mark scheme opens with “*less water available in deserts*”, then wants:

a **longer loop of Henle** relative to kidney size and a **thicker medulla** → **more ions pumped out** → a **steeper, deeper water potential gradient** → **more water reabsorbed** from the collecting duct → a **small volume of very concentrated urine, conserving water**.

10.5 Osmoregulation and ADH — the 4-mark chain

Mar 2022 Q4c:

• “*water potential of blood monitored by osmoreceptors in hypothalamus*” • “*release of ADH from (hypothalamus) pituitary gland*” • “*(in blood to kidneys) where ADH acts on collecting ducts / distal convoluted tubule*” • “*increase in permeability to water of cell membranes*” — “*ACCEPT insertion of aquaporins into cell membranes (of CT and DCT)*”

Then: **more water is reabsorbed by osmosis**, giving a **small volume of concentrated urine**; the **water potential of the blood rises**, detected by the osmoreceptors, so **ADH secretion decreases** — **negative feedback**.

The converse: high blood water potential → **less ADH** → collecting ducts **less permeable** → **large volume of dilute urine**.

Applications: **alcohol inhibits ADH** → dilute urine → dehydration. **Diabetes insipidus** — no ADH → copious dilute urine. **MDMA causes excess ADH** → water retention → dangerously **dilute blood**.

Urine volume calculation (Jun 2021 Q4e) — the mark scheme’s two steps:

$$0.83 \times 60 = 49.8 \text{ (cm}^3 \text{ per hour)}$$
$$\times 24, \text{ converted and given to 2 s.f.} = 1.2 \text{ dm}^3 \text{ per day}$$

10.6 Producing urea

Excess amino acids cannot be stored. In the **liver** they are **deaminated** — the **amino group is removed**, forming **ammonia** and an organic acid. The organic acid enters respiration; **ammonia is highly toxic**, so it is converted to the **less toxic urea**. Urea is **transported in the blood plasma to the kidney** and removed by **ultrafiltration**.

11. Transcription factors, peptide and steroid hormones

🔑 **Transcription factor:** a protein that **binds to a specific DNA sequence (the promoter)** and either **allows or prevents RNA polymerase binding**, so switching the gene on or off.

The signalling chain — Oct 2019 Q7h, the mark scheme's four points:

1. "(disease) cells produce **signal molecules**" — "ACCEPT **chemicals / hormones / cytokines / metabolites / enzymes**"
2. "(these) signal molecules **bind to receptors on cells**" — "ACCEPT **bind to transcription factors in the cytoplasm/nucleus**"
3. "idea of **triggering signal pathways**" — "e.g. **activate second messengers**"
4. "**changing transcription factor activity / concentrations**" — "ACCEPT **forming or activating a transcription factor**"

The two mechanisms — a guaranteed comparison:

	Peptide hormones (insulin, ADH, glucagon, adrenaline)	Steroid hormones (oestrogen, testosterone)
Solubility	Not lipid-soluble	Lipid-soluble
Cross the membrane?	No	Yes — diffuse straight through
Receptor	On the cell surface membrane	In the cytoplasm/nucleus
Mechanism	Activates a second messenger (cAMP) → cascade → activates transcription factors	The hormone–receptor complex enters the nucleus and itself acts as a transcription factor
Speed / duration	Fast, short-lived	Slower, longer-lasting
Described as acting	Extracellularly	Intracellularly

Why a second messenger is needed: the peptide hormone **cannot enter the cell**, so the signal must be **relayed** — and the cascade **amplifies** it.

PART B — TOPIC 8: COORDINATION, RESPONSE AND GENE TECHNOLOGY

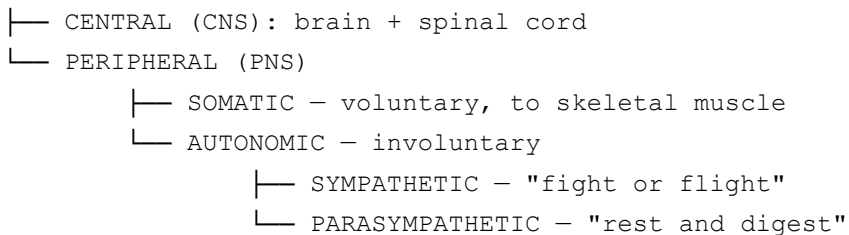
12. Neurones and the nervous system

Type	Direction	Structure
Sensory	Receptor → CNS	Long dendron, short axon; cell body on a side branch (dorsal root ganglion)
Relay	Within the CNS	Short , highly branched
Motor	CNS → effector	Cell body in the CNS , very long axon

Myelination: Schwann cells wrap repeatedly around the axon; their membranes form the **myelin sheath**, an **electrical insulator**. Gaps between adjacent Schwann cells are the **nodes of Ranvier**, where **voltage-gated channels are concentrated**.

Jan 2021 Q6a(i): the cell that forms the myelin sheath is the **Schwann cell**. Q6a(ii): impulse conduction is "**faster at nodes of Ranvier**."

NERVOUS SYSTEM



The general pathway:

STIMULUS → RECEPTOR → SENSORY neurone → CNS → MOTOR neurone → EFFECTOR → RESPONSE

Receptors are transducers — they **convert one form of energy into an electrical impulse** (a generator potential).

The spinal reflex arc: receptor → sensory neurone entering by the **dorsal root** → **relay neurone in the grey matter** → motor neurone leaving by the **ventral root** → effector. **Grey matter = cell bodies, dendrites and synapses** (unmyelinated). **White matter = myelinated axons** (white from the lipid). **Why reflexes are advantageous:** **fast** (few synapses; no conscious processing), **automatic** and **protective**. Impulses also travel **to the brain**, so awareness follows **after** the response.

13. The resting potential and the action potential

13.1 The resting potential — the 4-mark chain

Jan 2021 Q6b, the mark scheme's points:

• *"active transport by sodium potassium (ion) pump"* • *"of sodium ions out of the axon and potassium ions into the axon"* • *"(passive) diffusion of potassium ions out of the axon"* — *"ALLOW passive diffusion described"* • *"so that the inside of the membrane is negatively charged"* — *"ALLOW axon for membrane"*

The full picture: the pump moves **3 Na⁺ out for every 2 K⁺ in**, using **ATP**; the membrane is **more permeable to K⁺** (potassium leak channels), so K⁺ **diffuses back out**; large **negatively charged proteins** are trapped inside. Resting potential ≈ -70 mV.

13.2 The action potential — the five stages

1. **Stimulus:** some voltage-gated **Na⁺ channels open**; Na⁺ enters.
2. **Threshold (-55 mV):** many more **Na⁺ channels open** — **positive feedback**.
3. **Depolarisation:** **Na⁺ floods in**; the inside becomes positive, reaching about **+40 mV**.
4. **Repolarisation:** **Na⁺ channels close**, **voltage-gated K⁺ channels open**, **K⁺ diffuses out**.
5. **Hyperpolarisation:** K⁺ channels are **slow to close**, so the potential **overshoots to ~ -80 mV**; the **Na⁺/K⁺ pump** restores the resting potential.

Refractory period: the **Na⁺ channels are inactivated and cannot reopen**. Three consequences: the impulse travels **in one direction only**; action potentials remain **discrete**; the **maximum frequency is limited**.

All-or-nothing: an action potential occurs **only if the threshold is reached**, and is **always the same size**. **Stimulus intensity is coded by the frequency** of action potentials and by the **number of neurones** recruited.

Speed depends on: **myelination**, **axon diameter** (wider = faster, less resistance to ion flow) and **temperature**.

13.3 Saltatory conduction — the mark scheme's own wording

Jun 2021 Q3a:

*"(depolarisation) **only occurs at nodes of Ranvier**" "and therefore the nerve impulse*

'jumps' from one node to the next which is known as saltatory conduction" — "Accept the influx of sodium ions at one node creates enough depolarization to reach the threshold of the next."

Why it is advantageous: much faster (up to $\sim 100 \text{ m s}^{-1}$ versus $\sim 1 \text{ m s}^{-1}$), and **more energy-efficient** because **fewer ions cross the membrane**, so **less ATP** is needed by the pump.

Demyelination (Jun 2021 Q3b, 2 marks) — the mark scheme:

*• "when the myelin sheath is destroyed, **transmission of the nerve impulse is impaired**" — "slower transmission of impulses" • "**fewer nerve impulses reach the muscle**"*

Consequences: **muscle weakness, loss of coordination, numbness, visual disturbance** (multiple sclerosis, ADEM).

14. Synapses

14.1 The presynaptic events — 3 marks, verbatim

Jun 2019 Q7g:

1. *"**calcium ions (entering presynaptic neurone) bind to vesicles containing neurotransmitter**"*
2. *"causing vesicles to **move to / fuse with the presynaptic membrane**"*
3. *"**releasing contents into the synapse**" — "ALLOW release contents by **exocytosis**"*

14.2 The full chain

1. An **action potential arrives** at the presynaptic membrane.
2. **Voltage-gated calcium channels open; Ca^{2+} diffuses in.**
3. **Vesicles of neurotransmitter move to and fuse with** the presynaptic membrane.
4. Neurotransmitter is released **by exocytosis** into the **synaptic cleft**.
5. It **diffuses across** and **binds to specific receptors on the postsynaptic membrane**.
6. This **opens sodium (cation) channels; Na^+ enters**, causing **depolarisation**.
7. If the **threshold** is reached, a **new action potential** is generated.

Removal: acetylcholinesterase hydrolyses acetylcholine into choline and ethanoic acid, which are reabsorbed and recombined using ATP — preventing continuous stimulation.

Why synapses exist: unidirectionality (transmitter only presynaptic, receptors only postsynaptic) · summation (temporal and spatial) · filtering of low-level stimuli · inhibition · roles in learning and memory.

Blocking vesicle fusion (Jan 2021 Q6c) — the mark scheme:

"no fusion of (secretory) vesicles with the pre-synaptic membrane" — "ALLOW fewer vesicles fuse" — or "changes to pre-synaptic / vesicle membrane proteins" which "prevents fusion of vesicles with the pre-synaptic membrane."

14.3 The pupil reflex

Bright light	Dim light
Circular muscles contract, radial relax	Radial muscles contract, circular relax
Pupil constricts — protects the retina	Pupil dilates — more light enters
Parasympathetic	Sympathetic

An **antagonistic** response of the iris, under **autonomic** control.

15. Drugs affecting synaptic transmission

15.1 Lidocaine — 3 marks, with one telling rejection

Jan 2021 Q6d:

• "binds to (voltage dependent) sodium ion channels" — "ALLOW blocks / inhibits" • "blocking the (rapid) diffusion of sodium ions into the axon" • "reducing depolarisation (of the membrane)" — "IGNORE no depolarisation" • "below the threshold to trigger an action potential"

△ **"No depolarisation occurs" is ignored.** You must say depolarisation is **reduced below the threshold.**

15.2 The five named drugs

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Drug	Mechanism	Effect
Nicotine	An agonist — similar shape to acetylcholine, so it binds to nicotinic ACh receptors and opens sodium channels , causing depolarisation. Not broken down by acetylcholinesterase , so the effect is prolonged. Also raises dopamine in the reward pathway	Stimulation; addiction
Lidocaine	Blocks voltage-gated sodium channels (see above)	Local anaesthetic
Cobra venom alpha toxin	An antagonist — binds irreversibly to acetylcholine receptors , blocking them so ACh cannot bind	Paralysis , including respiratory muscles
L-DOPA	Jan 2021 Q7a(i), the mark scheme's own answer: " L-DOPA crosses the blood brain barrier and is then converted to dopamine in the brain " (dopamine itself cannot cross)	Treats Parkinson's
MDMA (ecstasy)	Blocks the reuptake transporters for serotonin , so serotonin remains in the cleft longer	Elevated mood; long-term serotonin depletion ; excess ADH → hyponatraemia

△ Learn the three categories: *agonist* mimics and activates; *antagonist* blocks without activating; *reuptake inhibitor* **prolongs** the transmitter's action.

16. Rods, cones and vision

16.1 Rhodopsin

Rhodopsin = opsin (protein) + retinal (from vitamin A). Jun 2019 Q4a's mark scheme is precise: "*retinal is a part of rhodopsin the photoreceptor in mammals*", and "*rhodopsin is the photoreceptor in mammals*" — as opposed to **phytochrome** in plants.

In the DARK:

Sodium ions flow continuously into the rod through **open cation channels** — the **dark current**. The rod is **depolarised** and **continuously releases neurotransmitter**, which

INHIBITS the bipolar neurone. No impulse is sent.

In the LIGHT:

1. **Light is absorbed by rhodopsin;** the **retinal changes shape.**
2. Rhodopsin **splits into opsin and retinal — bleaching.**
3. The opsin triggers a cascade that **CLOSES the cation channels.**
4. **Na⁺ can no longer enter,** but is still **pumped out,** so the rod **HYPERPOLARISES.**
5. The rod **stops releasing neurotransmitter.**
6. The **inhibition of the bipolar neurone is removed,** so it **depolarises.**
7. An **action potential** is generated in the **optic (ganglion) neurone.**

△ **The counter-intuitive point that is repeatedly examined:** light makes the rod **hyperpolarise and stop releasing transmitter.** The signal is the **removal of inhibition.**

Recovery requires **resynthesis of rhodopsin using ATP** — hence **dark adaptation,** and why **vitamin A deficiency causes night blindness** (Oct 2021 Q2).

16.2 Rods versus cones

	Rods	Cones
Pigment	Rhodopsin (one type)	Iodopsin — three types (red, green, blue)
Colour vision	No	Yes
Sensitivity	High — work in dim light	Low — need bright light
Convergence	Many rods → one bipolar neurone	One cone → one bipolar neurone
Acuity	Low	High
Distribution	Peripheral; absent from the fovea	Concentrated at the fovea

Why rods work in dim light: because **many rods converge on one bipolar neurone,** their small effects **add together (spatial summation),** so the **threshold can be reached even in dim light.** **Why rods give low acuity:** the brain **cannot tell which individual rod was stimulated,** so two nearby points appear as one. **Why cones give high acuity:** each cone

has its **own bipolar and optic neurone**, so the brain receives a **separate signal from each**.

Nocturnal adaptations (meerkat, tarsier, harbour seal): **high proportion of rods, large pupil and eye**, and often a **tapetum lucidum** reflecting light **back through the retina** for a second chance at absorption.

Reading absorption spectra (Jun 2019 Q4b): the mark scheme's rejections make the point — " $0.82 - 0.56 = 0.26 \text{ a.u.}$ **not** 0.26 nm ", and "**70 is the difference in wavelength, not the difference in absorbance.**" **Read which axis the question asks about, and use the right units.**

17. Habituation

📌 **The mark scheme's definition (Oct 2019 Q2a(iv))**: *"habituation is a form of learning in which an organism decreases or ceases its responses to a stimulus."*

The mechanism: with repeated stimulation, **less calcium enters the presynaptic knob**, so **less neurotransmitter is released**, so the **threshold of the postsynaptic neurone is not reached** and **no action potential** is generated.

The advantage: the organism **stops responding to a repeated harmless stimulus**, so it **does not waste energy** and remains **able to respond to new or threatening stimuli**.

Investigating it (Jun 2021 Q5, leech; Jan 2026 Q5, maze learning): stimulate at **fixed intervals** and record the **magnitude or presence of the response**; control the **intensity and interval**; use several organisms and **average**; show **recovery after a rest period** to prove the animal is not simply exhausted or injured.

Designing a learning experiment (Jun 2019 Q7j, 4 marks) — the mark scheme's structure:

• "allow animal to be **stung**" • "while letting them **hear the honeybee**" — "ALLOW so that the **sound of the bee is associated with a sting**" • "(at time intervals) after being stung, **observe response to the sound of honeybees**" — "e.g. run away / goes down burrow" • "idea that **if learning takes place the avoidance response is quicker or stronger**"

That template — expose, pair the stimulus, test later, state the expected change — answers any "design an investigation into learning" question.

18. The brain and imaging

18.1 The regions — the mark scheme's own labelling

Oct 2019 Q2a(iii): "*P is the cerebral hemisphere, Q is the cerebellum, R is the medulla oblongata, S is the hypothalamus.*" And Q2a(iv): "*homeostasis... covers temperature regulation*", with the rejection "*respiration does not regulate temperature.*"

Region	Functions
Cerebral hemispheres	Conscious thought , decision-making, personality , memory , language; receives sensory information and initiates voluntary movement
Hypothalamus	Homeostatic control — thermoregulation , osmoregulation , hunger and thirst; controls the pituitary ; produces ADH
Pituitary gland	Secretes hormones; the posterior lobe releases ADH
Cerebellum	Coordination of movement, balance and posture; fine motor control
Medulla oblongata	Non-conscious control — heart rate and ventilation rate , blood pressure, swallowing

18.2 Imaging

Technique	How	Shows	Advantages	Disadvantages
CT	X-rays from many angles	Structure	Quick; good for bleeds and tumours	Ionising radiation ; poorer soft-tissue detail
MRI	Magnetic field + radio waves ; detects hydrogen nuclei	Structure , excellent soft tissue	No ionising radiation ; high resolution	Expensive; noisy ; unsuitable with metal implants
fMRI	MRI detecting blood oxygenation changes	Brain ACTIVITY in near real-time	Non-invasive; no tracer	Expensive; subject must stay still ; measures blood flow, an indirect proxy
PET	Radioactive tracer emitting	Metabolic	Shows function ; detects disease	Radioactive isotope ;

	positrons	activity/function	before structural change	expensive
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How the mark scheme frames an fMRI answer (Jun 2019 Q7i, 2 marks):

• *"using fMRI"* • *"compare images from control and stung individuals / observe changes in the stung individual"*

And the study-design version (Oct 2019 Q2b): *"idea of **simulating a game of squash**"* — *"e.g. **show a video** of someone hitting a squash ball / **ask someone to think about** hitting a squash ball"* — then *"**stimulates parts of brain**"*.

⚠ **The exam framing:** match the technique to whether the question is about **structure** (CT, MRI) or **function/activity** (fMRI, PET), and whether **radiation exposure** matters.

19. Neurotransmitter imbalance and disease

19.1 Parkinson's — the 3-mark chain

Jan 2021 Q7a(ii), the mark scheme's four points:

• *"fewer dopamine molecules to bind to the ligand gated sodium channels / receptors"* • *"on the post-synaptic membrane"* • *"initiating fewer action potentials in (the post-synaptic neurone)"* • *"fewer impulses sent to parts of the brain controlling motor (function)"*

The cause: dopamine-secreting neurones in the substantia nigra die, so less dopamine is available. **Symptoms:** tremor at rest, muscle rigidity, slowness of movement (**bradykinesia**), postural instability. **Treatments:** L-DOPA • dopamine agonists • MAO-B inhibitors (slow dopamine breakdown) • **deep brain stimulation** • gene and stem cell therapy.

19.2 Depression — serotonin

Associated with a **low serotonin concentration**. SSRIs block the reuptake transporters, so **serotonin remains in the cleft longer** and **continues to bind to postsynaptic receptors**.

How serotonin acts (Jun 2019 Q7h, 3 marks):

"serotonin is a neurotransmitter" • *"binds to pain receptors"* — *"ALLOW binds to post-synaptic receptors/membrane"* • *"on sensory neurones"* • *"impulses pass to*

20. Plant responses

20.1 Phytochrome — and the photoreceptor MCQ

Jun 2019 Q4a: the plant photoreceptor is *phytochrome*. The mark scheme rejects the others precisely:

"IAA (auxin) is a plant hormone produced in response to light stimulation but is not itself a photoreceptor" "retinal is a part of rhodopsin, the photoreceptor in mammals"

Pr —absorbs RED (660 nm)→ Pfr Pfr is the ACTIVE form
Pfr —absorbs FAR-RED (730 nm)→ Pr
Pfr —slowly, in DARKNESS→ Pr

In daylight (more red than far red), **Pfr accumulates**. **In darkness**, Pfr **slowly reverts to Pr**, so the **length of the night determines how much Pfr remains** — this is how a plant measures day length.

The molecular action: Pfr acts as, or activates, a transcription factor, binding to the **promoter** of specific genes so they are **transcribed**, bringing about **flowering, germination or stem elongation**.

Photoperiodism: long-day plants flower when **Pfr is high**; **short-day plants** flower when **Pfr is low**.

△ **The red/far-red flash experiment (Jun 2026 Q7) — the last flash determines the response:**

Dark only	→ Pfr low	→ 4 % germinated
Red, dark	→ Pfr HIGH	→ 88 %
Red, far-red, dark	→ Pfr low	→ 2 %
Red, far-red, red, dark	→ Pfr HIGH	→ 98 %
Red, far-red, red, far-red	→ Pfr low	→ 4 %

Conclusion: germination requires **Pfr**, and the effect of red light is **reversed by far-red**.

20.2 Auxin (IAA)

Produced in the **shoot and root tips**. **Mechanism:** IAA causes **H⁺ to be pumped into the**

cell wall, lowering the pH, which activates expansins that loosen the cross-links between cellulose microfibrils, so the wall becomes flexible and the cell elongates as water enters. IAA also affects transcription.

Phototropism: light causes IAA to move to the shaded side; the higher IAA concentration causes greater cell elongation there, so the shoot bends towards the light.

△ In roots IAA INHIBITS elongation, so roots show positive gravitropism. **Apical dominance:** IAA from the apical bud inhibits lateral bud growth.

20.3 Gibberellin and the germinating cereal grain

1. Water is absorbed and the embryo produces gibberellin.
2. Gibberellin diffuses to the aleurone layer.
3. It stimulates transcription of the gene for amylase.
4. Amylase hydrolyses the starch in the endosperm into maltose, then glucose.
5. Glucose is absorbed by the embryo for respiration and to make cellulose.

The practical: soak barley grains; sterilise the surfaces with dilute bleach (so surface microorganisms do not digest the starch themselves); cut in half; place cut-side down on starch agar — with embryo, without embryo, without embryo + gibberellin, without embryo + water (control); incubate; flood with iodine. **Result:** starch agar stains blue-black; a clear zone shows amylase has hydrolysed the starch. **With embryo → clear zone. Without embryo + water → none. Without embryo + gibberellin → clear zone.**
Conclusion: the embryo produces gibberellin, which induces amylase synthesis.
Measure the clear-zone diameter in two directions and average, or the area.

20.4 Nervous versus hormonal coordination — the table question

Jun 2019 Q3c, 3 marks. The mark scheme's five paired rows:

Hormonal	Nervous
use chemicals	use impulses — "ALLOW both use chemicals"; "ALLOW use electrical signals"
transported in blood	carried by neurones
slow	fast

long-lasting response	short-lived response
widespread effects	localised effects

⚠ You must give both halves of each pair — a one-sided statement scores nothing.

21. Gene technology

21.1 Recombinant DNA

1. **Restriction endonucleases** cut DNA at a **specific recognition sequence**; a **staggered cut** leaves **sticky ends**.
2. The **plasmid is cut with the SAME restriction enzyme**, so both have **complementary sticky ends**.
3. The sticky ends **anneal by complementary base pairing**.
4. **DNA ligase joins the sugar–phosphate backbones**, forming **phosphodiester bonds**.

Alternatively the gene is made as **cDNA from mRNA using reverse transcriptase**, or synthesised in a **gene machine**.

⚠ **Both halves matter:** the **same** restriction enzyme, and **DNA ligase** forming **phosphodiester bonds**.

Getting it into cells: heat shock with calcium ions · electroporation · viral vectors · liposomes · gene gun (plants) · *Agrobacterium tumefaciens* (plants) · microinjection. **Marker genes** (antibiotic resistance, **GFP**) allow **transformed cells to be identified and selected**.

Products: human insulin, growth hormone, factor VIII, vaccines; GM plants (**golden rice**, Bt crops); GM animals ("pharming" — proteins in milk). **Advantages over animal-derived products:** identical to the human protein, so less likely to cause an immune reaction; no disease transmission; large quantities cheaply; avoids ethical objections.

21.2 Microarrays and bioinformatics — the 4-mark chain

Jan 2021 Q7c, the mark scheme's five points:

• "microarrays allow identification of active genes / gene transcription" • "the

activity of many genes can be analysed in a single sample" · "by collecting information about genetic differences from many individuals (with or without Parkinson's)" · "bioinformatics / computers / databases / algorithms used to analyse the data" — "ALLOW develop algorithms to identify genomes / gene sequences" · "(key) differences between healthy and diseased individuals can be identified"

The method behind it:

A microarray carries **thousands of spots of single-stranded DNA probes** for known genes in a **known grid position**. **mRNA is extracted** (present only for genes being **transcribed**), converted to **labelled cDNA** using **reverse transcriptase**, and **washed over the array**. Complementary sequences **hybridise**; unbound cDNA is **washed off**; the array is **scanned** and **fluorescent spots show which genes were expressed**. Two-colour labelling compares **healthy versus diseased** samples directly.

🔗 **Bioinformatics:** the use of **computers, algorithms and databases** to **store, organise and analyse very large quantities of biological data** — genome sequences, protein structures and gene-expression data — because the volume is **far too large to analyse by hand**.

21.3 Risks and benefits of GMOs

Benefits: higher **yields** and **food security** · **pest resistance** → **less pesticide** · **herbicide, drought and disease resistance** · **improved nutrition** (golden rice) · **longer shelf life** · **cheaper, purer medicines** · **potential gene therapy**.

Risks: **gene transfer to wild relatives** → "superweeds" · harm to **non-target organisms**, reducing **biodiversity** · **monoculture** · possible **allergenicity** · **antibiotic-resistance marker genes** transferring to bacteria · **unknown long-term effects** · farmers **dependent on seed companies** · **ethical and religious objections** · questions of **regulation and labelling**.

A good conclusion: risks and benefits differ for each organism and each use, so each case should be **assessed individually**, with **testing, regulation, monitoring and clear labelling**.

21.4 Stem cells and inheritance — the synoptic edges

Stem cell risks (Jan 2021 Q5): "*could continue to divide*" — "ALLOW could form a **tumour / cancer**"; "*stem cells could differentiate into other cell types*."

Recessive inheritance (Oct 2019 Q7i): "*a recessive trait requires two copies of the mutation / no copies of the functioning gene*" — "ACCEPT recessive trait **expressed in homozygous individuals**"; "ACCEPT '**allele**' for '**mutation**'."

Evolution of resistance (Jun 2019 Q7f) — the mark scheme's four points, which mirror Unit 4 exactly:

"mutation (in the cockroach)" · "allele for resistance present in the population" · "selective pressure acts on the population" — "ALLOW natural selection takes place" · "resulting in a change in allele frequency" · "allele conferring resistance / selective pressure not present for the other species"

22. Development and critical windows

A recurring context (Jun 2019 Q6, Mar 2021 Q5, Jan 2024 Q3).

🔗 The mark scheme's definition of a critical window (Jun 2019 Q6a(i)):

"a period of time during which a stimulus is required for (normal) development" · "period when rats must be exposed to low oxygen concentration" · "for development of the ventilation centre"

How to answer the data part (Q6a(ii)): compare the groups directly — *"the group at 11 to 15 days showed a smaller increase in breathing rate"* — **quoting figures**, then explain in terms of the **critical window having been missed**.

PART C — THE ARTICLE QUESTION AND EXAM TECHNIQUE

23. Preparing the pre-released article — the 20-mark plan

This is the highest-value preparation available. Start the day the article is released.

Step 1 — Read it three times, differently

1. **Straight through**, for the argument. Do not stop.
2. **With a dictionary and this note**, writing a **definition of every unfamiliar term**. Vocabulary is worth 4–6 marks directly.
3. **Paragraph by paragraph**, annotating your own copy.

Step 2 — Map every paragraph to a section of this note

This is the step that converts the marks. Two worked examples:

From the Jan 2025 hydration article:

Paragraph theme	Maps to
Dehydration and thirst	§10.5 ADH and osmoregulation, §9 negative feedback
Electrolytes in sports drinks	§10.3 selective reabsorption , water potential
Isotonic vs hypotonic drinks	Osmosis (Unit 1 synoptic)
Older people and dehydration	§9 homeostasis , reduced ADH sensitivity
Kidney and urine colour	§10.4, §10.5

From the Oct 2022 Nobel Prize article (TRPV1, TRPM8, Piezo):

Paragraph theme	Maps to
Temperature-sensitive ion channels	§12 receptors as transducers , generator potentials
"A measurable electric current in a cell line"	§13 membrane potential and depolarisation
Mechanically activated channels	§12–13 stimulus → receptor → impulse
Pain pathways	§14 synapses, §19.2 serotonin and pain receptors

From the Jun 2019 *Venom* article — and this is confirmed by the mark scheme itself: the questions were on **calcium ions and vesicle fusion (§14.1)**, **serotonin binding to pain receptors on sensory neurones (§19.2)**, **fMRI comparison of control and stung individuals (§18.2)** and **designing a learning experiment (§17)**. Every one of them is in this note.

Step 3 — Prepare the five predictable question types

Type	Preparation
Define a term from the article	A glossary of every technical term
	For each paragraph, write the underlying mechanism in full —

Explain the biology behind a statement	from this note
Interpret a study described	Note the method, sample size, controls and conclusions of each study mentioned
Evaluate the evidence	For each study, note limitations : sample size, correlation vs causation, controls, generalisability, funding
Suggest applications / further work	Note what the research could lead to and what remains unknown

Step 4 — Practise on old article questions

Work through article questions from **previous sessions even though you have not read those articles**. The question *shapes* are identical every year, and you will see how much of each answer comes from the specification rather than the text.

Step 5 — In the exam

- **Skim the clean copy** before starting, but do not re-read it fully.
- **Questions usually follow the article in order** — use the paragraph numbers to navigate.
- When a question says “*using the information in the article*”, you **must refer to it**; when it does not, answer from **this note**.
- **Budget 22–25 minutes**. Do not let Questions 1–7 eat into it.

24. Levels-based marking and QWC

Level	Marks	Descriptor
1	1–2	Isolated elements ; generalised, vague statements; simple links
2	3–4	Some accurate knowledge applied to the context; some logical linkages ; reasoning partially developed
3	5–6	Accurate and detailed knowledge applied throughout; coherent, logically structured , with sustained linkages

Older papers add **QWC** — “*emphasis is clarity of expression*” (Jun 2019 Q5b(i)).

⚠ **A list of correct facts caps at Level 2.** Level 3 requires **linked reasoning applied to the specific context or data** — “in this patient”, “because the loop is longer”, “since the RQ fell to 0.7”.

For a data-based * question, alternate description and explanation, as the June 2019 lactate scheme does (§2.3): describe the trend **with figures**, then explain it, then the next trend, then its explanation.

Connectives are the mechanism: *because · so · this means that · therefore · as a result · which leads to · eventually.*

25. Command words

Command	What is required
State / Give / Name	A short fact; no explanation
Describe	What happens or what the data show , with figures
Explain	Reasons — must contain <i>because / so / due to</i>
Suggest	Apply biology to an unfamiliar context; more than one answer usually accepted
Calculate	Working, unit , significant figures , correct rounding
Deduce	A conclusion from the data with the reasoning
Compare / Contrast	Comparative language for both items
Comment on	A judgement supported by evidence
Discuss / Evaluate	Both sides , then a conclusion
Devise / Design	IV, DV, controlled variables, repeats and a control

26. Calculations that recur

cardiac output = heart rate $\times$ stroke volume
 heart rate = $60 \div (\text{time for one cardiac cycle})$
 minute ventilation = tidal volume $\times$ breathing rate
 RQ = $\text{CO}_2 \text{ produced} \div \text{O}_2 \text{ consumed}$ (no units)
 respirometer = distance moved $\times \pi r^2 \div \text{time} \div \text{mass}$
 magnification = image $\div$ actual (convert mm $\rightarrow$ μm)
 % change = $(\text{change} \div \text{original}) \times 100$

Worked from the mark schemes themselves:

- **ECG (Jun 2019 Q1b(i)):** $(11.7 \times 0.2) \div 3 = 0.78 \text{ s per beat} \rightarrow 76.9 \text{ bpm}$ — *"ALLOW 75.9 to 77."*
- **Urine volume (Jun 2021 Q4e):** $0.83 \times 60 = 49.8$ then $\times 24 \rightarrow 1.2 \text{ dm}^3$, to 2 s.f.
- **Absorbance (Jun 2019 Q4b):** $0.82 - 0.56 = 0.26 \text{ a.u.}$ — **not nm**, and **70 nm is the wavelength difference**, not the absorbance difference.
- **RQ (Jan 2024 Q1):** $18 \div 25.5 = 0.71$.

Ranges are given for graph read-offs — take values carefully and show the numbers you read.

27. Mark-scheme phrase bank

Respiration

- **"substrate level phosphorylation"** — *"ALLOW some ATP is also produced directly"*
- **"reference to chemiosmosis"** · *"protons moved into the intermembrane space" · "ATP synthase / stalked particles"*
- **"formation of phosphate bond between phosphate in ADP and inorganic phosphate"** — not *"phosphorylation makes ATP"*
- **"H⁺ / hydrogen ion / proton"** — not *"hydrogen"*
- **"it is NAD not reduced NAD that accepts electrons"**
- **"switching to lipid being used rather than carbohydrate"**

Muscle and heart

- **"the structure labelled X is made from myosin"; "W, X and Y do not change in length"**

- **"contraction initiated in a muscle cell"** (myogenic)
- **"produces waves of excitation / waves of depolarisation"** — not "nerve impulses"
- **"cardiac output is the volume of blood ejected from the left ventricle in each beat multiplied by the heart rate"**

Homeostasis and kidney

- **"the effectors reverse the detected change"**
- **"thermoreceptors detect an increase in core temperature" → "hypothalamus / thermoregulatory centre" → "effector organs" → "returns body temperature to set value"**
- **"forced out by high pressure" · "afferent arteriole greater diameter than efferent" · "pores in the basement membrane"**
- **"sodium ions actively transported out of the ascending limb" · "flow of filtrate in opposite directions"**
- **"osmoreceptors in hypothalamus" · "ADH from the pituitary" · "acts on collecting ducts / DCT" · "insertion of aquaporins"**

Nerves

- **"sodium potassium pump" · "passive diffusion of potassium ions out of the axon" · "inside of the axon is negatively charged"**
- **"only occurs at nodes of Ranvier" · "the impulse 'jumps' from one node to the next — saltatory conduction"**
- **"calcium ions bind to vesicles" · "vesicles fuse with the presynaptic membrane" · "released by exocytosis"**
- **"reducing depolarisation below the threshold" — not "no depolarisation"**
- **"L-DOPA crosses the blood brain barrier and is converted to dopamine in the brain"**
- **"fewer dopamine molecules to bind to receptors on the post-synaptic membrane" → "fewer action potentials"**

Coordination and gene technology

- **"hormonal: chemicals, blood, slow, long-lasting, widespread; nervous: impulses, neurones, fast, short-lived, localised"**
- **"phytochrome is the plant photoreceptor; rhodopsin is the photoreceptor in**

mammals"

- **"microarrays allow identification of active genes" · "bioinformatics used to analyse the data"**
 - **"changing transcription factor activity or concentrations"**
-

28. The twenty most commonly lost marks — each a mark-scheme rejection

1. **"Nerve impulses" for the SAN** — *IGNORE*; say **waves of excitation/depolarisation**.
2. **"Phosphorylation makes ATP"** — *IGNORE*; say a **bond forms between ADP and inorganic phosphate**, and name **ATP synthase**.
3. **"Hydrogen" instead of H⁺/proton** in the ETC — *IGNORE* *hydrogen / H₂ / H*.
4. **"Artery" instead of "arteriole"** in ultrafiltration — *ALLOW* *arteriole NOT* *artery*.
5. **"Medulla oblongata" for thermoregulation** — *IGNORE*; it is the **hypothalamus**.
6. **"No depolarisation" for lidocaine** — *IGNORE*; say **depolarisation is reduced below the threshold**.
7. **Saying the A band shortens** — *W, X and Y do not change in length*.
8. **Saying reduced NAD accepts electrons** — it is **NAD** that accepts them.
9. **Explaining a falling RQ by anaerobic respiration** — *ignore*; it means a **switch to lipid**.
10. **Saying light depolarises the rod** — light causes **hyperpolarisation** and **removal of inhibition**.
11. **Giving only one side of the nervous/hormonal table** — both halves of each pair are needed.
12. **Omitting that the ascending limb is impermeable to water**.
13. **Saying ADH "makes the kidney absorb more water"** without naming the **collecting duct/DCT** and **aquaporins**.
14. **Omitting the figures** in selective reabsorption answers — the scheme wants *"100 % of glucose", "70× increase for urea"*.
15. **Not stating that lactate fermentation regenerates NAD** — the whole point of it.

16. **Forgetting the control tube of glass beads** in the respirometer, and what it compensates for.
17. **Not stating that soda lime absorbs CO₂**, which is why the volume change measures oxygen only.
18. **Confusing fMRI (activity) with MRI (structure)**, or PET with CT.
19. **Quoting the wrong axis or unit** — “0.26 a.u. not 0.26 nm”.
20. **Under-preparing the article** — 20 marks, entirely predictable in shape.

29. Per-paper map

Paper	Questions 1–7	Article
2019 Jun	Heart, ECG, cardiac output · ACL and tendon grafts · homeostasis, thermoregulation, nervous vs hormonal · plants and light, phytochrome · respiration, lactate*, oxygen debt · critical window in development	<i>Venom</i>
2019 Oct	Plants and light · exercise coordination · mitochondria, ETC, chemiosmosis · joints · cardiac output and circulation · the eye · brain regions, habituation, fMRI	<i>First in Fly: Drosophila</i>
2020 Oct	Anaerobic and aerobic respiration · kidney · MDMA · swimming · the eye · predatory toxins · Parkinson's	<i>National Geographic</i>
2021 Mar	Action potentials · muscle strength training · brain · sarcomere and myosin · RPE65 blindness · exercise and cardiac output · Parkinson's	<i>Education in Chemistry</i>
2021 Apr	Four stages of respiration · nervous response · myelination and saltatory conduction · kidney · leech habituation · SAN/AVN/bundle of His · light responses	<i>Journal of...</i>
2021 Oct	Skeletal muscle fibres · vitamin A · retina · homeostasis · aerobic and anaerobic ATP · T lymphocytes · rats, drugs and stress	<i>Rethinking Alzheimer's</i>
2022 Jan	Mitochondrion EM · muscles, bones, joints · whales · kidney filtration and fluid balance · light responses · nervous system · alpha-1 antitrypsin	<i>Science Direct</i>
2022	Brain neurones · kidney · chemical regulation in plants and animals ·	

Jun	Parkinson's and substantia nigra · nervous responses · internal and external environment · mitochondria	<i>StatPearls</i>
2022 Oct	Mitochondria · giraffe neck · relay neurone · resting heart rates · camel · plant responses · GM animals and plants	<i>Nobel Prize: TRPV1, TRPM8, Piezo</i>
2023 Jan	Elbow joint · brain imaging · nervous communication and ions · cancer, mutation, epigenetics · aerobic respiration · slow and fast twitch fibres · rats and sucrose	Article
2023 Jun	Eye and light intensity · ATP in metabolism · ventilation rate, RQ · muscles and movement · kidney and osmoregulation · ion homeostasis · serotonin	Article
2023 Oct	Recombinant DNA · meerkat eyes · ADEM and myelin · resting heart rates · dopamine, serotonin, acetylcholine · chemical control · antagonistic muscles	<i>New Scientist</i>
2024 Jan	Respiration and RQ from an equation · action potentials · <i>C. elegans</i> · nervous responses · DNA variation and gene activation · Usain Bolt · kangaroo rat	<i>Clinical Nutrition</i>
2024 Jun	Muscles, tendons, ligaments · brain functions · environmental responses · the sarcomere · Chamorro people and flying foxes · anaerobic and aerobic ATP · kidney reabsorption	<i>New Scientist</i>
2024 Oct	Neurones, receptors, effectors · chemical coordination · muscle fatigue · Parkinson's · pronghorns · kidney and plasma concentration · deficiency diseases	<i>Alzheimer's</i>
2025 Jan	Proteins · brain cross-section · kidney filtration · sodium and potassium ions · insulin as a peptide hormone · neurotransmitters at synapses · mitochondrion	<i>New Scientist: hydration</i>
2025 Jun	Nervous coordination · respiration in mitochondria · drugs and the nervous system · Parkinson's and dopamine · exercise, heart rate, blood pressure · plant coordination · hormones	<i>Risk of heart disease</i>
2025 Oct	Anaerobic and aerobic respiration · thale cress seeds · kidney and toxins · harbour seals · the eye · coordination and learning · skeletal tissues	<i>Popular Science</i>
2026	Drugs and impulse transmission · RQ and substrate · spinal reflex	

Jan	arc · kidney and glucose reabsorption · maze learning · exercise and cardiac output · light-absorbing molecules · twitch fibres	<i>Nature</i>
2026 Jun	GM food crops · tarsier eyes · kidney · emperor penguins · muscles · nervous system · plant responses (red/far-red)	<i>New Scientist</i>

The architecture is fixed. Every paper has a **respiration** question, a **kidney** question, a **nervous system** question, a **muscle/movement** question, a **plant or animal response** question, usually an **eye** or **brain/drug** question, and the **20-mark article question**.

30. What an A* requires, and the final plan

The rule

A* requires **grade A for the IAL overall** (≥ 480 of 600 UMS) **and** $\geq 90\%$ of the UMS across the IA2 units combined (≥ 270 of 300).

The IA2 units are **4, 5 and 6** — 120 + 120 + 60 UMS. **The A* is decided entirely by A2 performance. Unit 6 is 60 of the 300**, and it is examined on the practical work behind Units 4 and 5 — so §6, §8.6 and §20.3 of this note are Unit 5 content **and** Unit 6 content.

Priority order

Priority	Content	Marks/paper
1	The article, mapped to this note (§23)	20
2	The four stages of respiration with locations, yields and the chemiosmosis chain (§2–§4)	10–18
3	Resting potential, action potential, saltatory conduction, synapse (§13, §14)	12–20
4	Kidney: ultrafiltration, PCT, loop of Henle, ADH — all four chains (§10)	8–15
5	The sliding filament theory and the sarcomere bands (§7)	4–8
6	Rods, cones and the rhodopsin mechanism (§16)	4–10
7	Phytochrome, IAA and gibberellin (§20)	4–10
8	Brain regions, imaging, Parkinson's and the five drugs (§15, §18,	5–14

	§19)	
9	The practicals (§6, §8.6, §20.3) — Unit 5 <i>and</i> Unit 6	5–10
10	Recombinant DNA, microarrays, GMO evaluation (§21)	4–12

Five-step routine for every question

1. Circle the COMMAND WORD and the MARK ALLOCATION
2. Underline the CONTEXT and any data — Level 3 answers use them
3. Calculations: working → answer → UNIT → significant figures
4. Explanations: write the number of statements the marks demand, LINKED, using the exact phrases in §27
5. Re-read: has it asked for a reason, a comparison, a conclusion or a limitation you have not given?

Exam-day tactics

- **Protect 22–25 minutes for Question 8.** Work backwards from the clock.
- **Bring the HB pencil** for graphs and diagram completion.
- On * questions, write **linked prose**, alternating **description with explanation** where there is data.
- **Quote figures with units** whenever describing data.
- Never leave a **“suggest”** question blank.
- Leave five minutes to check that every *explain* contains a **because**, every calculation has a **unit**, and every *evaluate* has a **conclusion**.